4-MONTH JOB CRACK CHALLENGE

For Data Scientist & Machine Learning Engineer

🧮 PART I: THE MATHEMATICAL & COMPUTATIONAL BEDROCK

Week 01: Linear Algebra & Vector Calculus for ML

- **Day 01** Vector Spaces, Subspaces, and Basis: The Geometry of Data.
- **Day 02** Linear Transformations, Matrices, and Systems of Equations.
- **Day 03** Eigenvalues, Eigenvectors, and Eigendecomposition.
- **Day 04** Singular Value Decomposition (SVD) and Principal Component Analysis Foundations.
- **Day 05** Multivariable Calculus: Partial Derivatives and the Gradient Vector.
- **Day 06** The Jacobian and Hessian Matrices: Understanding Curvature in Optimization.
- **Day 07** Matrix Calculus Notation Rigor & Backpropagation Foundations.

Week 02: Probability, Statistics & Bayesian Inference

- **Day 01** Probability Theory: Axioms, Conditional Probability, and Bayes' Theorem.
- **Day 02** Random Variables: Discrete vs. Continuous Distributions.
- **Day 03** Expectation, Variance, and Moments of a Distribution.
- **Day 04** Inferential Statistics: Central Limit Theorem and Sampling Distributions.
- **Day 05** Hypothesis Testing: P-values, Type I/II Errors, and Z/T-tests.
- **Day 06** Maximum Likelihood Estimation (MLE) vs. Maximum A Posteriori (MAP).

Day 07 Statistical Significance in A/B Testing for Product Data Science.

Week 03: Optimization & Linear Models

Day 01 Convex Optimization: Global vs. Local Minima and KKT Conditions.

Day 02 Simple Linear Regression: OLS Derivation & The Bias-Variance Decomposition Proof.

Day 03 Multiple Linear Regression and Multi-collinearity Diagnostics.

Day 04 Gradient Descent Variants: Batch, Stochastic (SGD), and Mini-batch.

Day 05 Regularization: Ridge (L2), Lasso (L1), and Elastic Net.

Day 06 Logistic Regression: The Sigmoid Function and Cross-Entropy Loss.

Day 07 Coding Workshop: Building Regression Models from Scratch in NumPy.

Week 04: Python for High-Performance Data Science

Day 01 Advanced Python: Generators, Decorators, and Context Managers.

Day 02 NumPy Mastery: Vectorization, Broadcasting, and Memory Efficiency.

Day 03 Pandas Deep Dive: Multi-indexing, GroupBy Mechanics, and Pivot Tables.

Day 04 Exploratory Data Analysis (EDA): Outliers and Distribution Shifts.

Day 05 Data Visualization Theory: Matplotlib, Seaborn, and Plotly.

Day 06 Data Cleaning Pipelines: Missingness Mechanisms and Categorical Encoding.

🏆 **Day 07** Milestone Project 01: End-to-End EDA and Statistical Report on a Real-World Dataset.

🤖 PART II: CORE MACHINE LEARNING & ALGORITHMS

Week 05: Supervised Learning I (Classification & Trees)

Day 01 K-Nearest Neighbors (KNN) and the Curse of Dimensionality.

Day 02 Decision Trees: Information Gain, Gini Impurity, and Entropy.

Day 03 Tree Pruning and Handling Overfitting.

Day 04 Support Vector Machines (SVM): Hyperplanes and the Kernel Trick.

Day 05 Naive Bayes: Geometric vs. Probabilistic Classification.

Day 06 Evaluation Metrics: Precision, Recall, F1-Score, ROC-AUC, and Model Calibration (Platt Scaling).

Day 07 Imbalanced Data Techniques: SMOTE, Weighted Loss, and Precision-Recall Curves.

Week 06: Supervised Learning II (Ensemble & Interpretability)

Day 01 Introduction to Ensemble Thinking: Wisdom of the Crowd.

Day 02 Bagging: Bootstrapping and Random Forests.

Day 03 Boosting Theory: AdaBoost and Weighted Weak Learners.

Day 04 Gradient Boosting Machines (GBM) and XGBoost Architecture.

Day 05 LightGBM and CatBoost: Handling Categorical Data and Execution Speed.

Day 06 Explainable AI (XAI): SHAP Values, LIME, and Global vs. Local Interpretability.

Day 07 Hyperparameter Tuning: GridSearch, RandomSearch, and Bayesian Optimization (Optuna).

Week 07: Unsupervised Learning & Dimensionality Reduction

Day 01 Clustering: K-Means, K-Medoids, and the Silhouette Score.

Day 02 Hierarchical Clustering and Dendrograms.

Day 03 Density-Based Clustering (DBSCAN) and Anomaly Detection.

Day 04 Dimensionality Reduction: PCA (Principal Component Analysis) Deep Dive.

Day 05 t-SNE and UMAP for High-Dimensional Visualization.

Day 06 Association Rule Learning: Apriori and FP-Growth.

🏆 **Day 07** Milestone Project 02: Customer Segmentation and Fraud Detection Pipeline.

Week 08: SQL, Big Data & Feature Engineering

Day 01 Advanced SQL: Joins, Subqueries, and CTEs (Common Table Expressions).

Day 02 Window Functions: RANK, LEAD/LAG, and Partitioning.

Day 03 Feature Engineering: Polynomial Features, Binning, and Interaction Terms.

Day 04 Scaling Data: StandardScaler vs. RobustScaler vs. MinMaxScaler.

Day 05 Working with NoSQL (MongoDB) and Large Scale Data (Spark Basics).

Day 06 Feature Selection: Recursive Feature Elimination (RFE) and Variance Inflation Factor (VIF).

Day 07 Data Leakage: How to Identify and Prevent It in Production Environments.

🧠 PART III: DEEP LEARNING & SPECIALIZED DOMAINS

Week 09: Neural Network Foundations & Scale

Day 01 The Biological Neuron vs. Artificial Perceptron.

Day 02 Multi-Layer Perceptron (MLP) and Architecture Design.

Day 03 Activation Functions: ReLU, Tanh, Softmax, and Swish.

Day 04 Deep Dive: Backpropagation and the Chain Rule in Code.

Day 05 Optimization for DL: Adam, RMSProp, and Momentum.

Day 06 Overfitting in DL: Dropout, Batch Normalization, and Early Stopping.

Day 07 PyTorch Basics & Distributed Machine Learning (Data Parallelism vs. Model Parallelism).

Week 10: Computer Vision (CV)

Day 01 Convolutional Neural Networks (CNN) Architecture.

Day 02 Pooling, Strides, and Padding Mechanics.

Day 03 Classic Architectures: LeNet, AlexNet, VGG.

Day 04 Modern CV: ResNet (Residual Connections) and Inception.

Day 05 Transfer Learning: Fine-tuning ImageNet Models.

Day 06 Object Detection: YOLO (You Only Look Once) and R-CNN Concepts.

Day 07 Image Augmentation Strategies for Small Datasets.

Week 11: Natural Language Processing (NLP) & GenAI

Day 01 Text Preprocessing: Tokenization, Stemming, and Lemmatization.

Day 02 Word Embeddings: Word2Vec, GloVe, and FastText.

Day 03 Sequence Modeling & Attention Mechanism: Why "Attention is All You Need."

Day 04 Transformers: BERT, GPT, and Encoder-Decoder Architectures.

Day 05 Large Language Models (LLMs) & Prompt Engineering Basics.

Day 06 Parameter-Efficient Fine-Tuning (PEFT): LoRA and QLoRA for LLMs.

Day 07 Retrieval-Augmented Generation (RAG) & Vector Databases (Pinecone/Milvus).

Week 12: Time Series & Recommendation Systems

Day 01 Time Series Basics: Trend, Seasonality, and Noise.

Day 02 Classical Models: AR, MA, ARIMA, and SARIMA.

Day 03 Modern Forecasting: Facebook Prophet and DeepAR.

Day 04 Recommendation Systems: Collaborative Filtering (User/Item-based).

Day 05 Matrix Factorization: ALS and SVD for Recommendations.

Day 06 Content-Based Filtering and Hybrid Systems.

🏆 **Day 07** Milestone Project 03: Build a Movie Recommender or Stock Price Predictor.

🚀 PART IV: MLOPS, SYSTEM DESIGN & CAREER CRACKING

Week 13: MLOps & Model Deployment

Day 01 The ML Lifecycle: Experiment Tracking with MLflow or W&B.

Day 02 Containerization: Docker for Data Scientists.

Day 03 Model APIs: Building REST APIs with FastAPI.

Day 04 Cloud Deployment: AWS (SageMaker) or GCP (Vertex AI) + Cloud Cost Optimization.

Day 05 CI/CD for ML: GitHub Actions and Automated Testing.

Day 06 Model Monitoring: Detecting Data Drift, Concept Drift, and System Failures.

Day 07 Scalable Serving: Kubernetes and TorchServe Basics.

Week 14: Data Structures, Algorithms & ML System Design

- **Day 01** Big O Notation and Time/Space Complexity.
- **Day 02** Arrays, Strings, and Hash Maps (Interview Essentials).
- **Day 03** Linked Lists, Stacks, Trees, and Queues.
- **Day 04** Sorting and Searching (Binary Search).
- **Day 05** Scalable System Architecture: Load Balancers, Latency vs. Throughput.
- **Day 06** ML System Design Case Studies: Netflix Recommendation Engine & Uber ETA Prediction.
- **Day 07** Coding Mock Interview: LeetCode Easy/Medium for Data Science.

Week 15: Capstone Project - The "Job-Winner" Portfolio

- **Day 01** Project Ideation: Solving a Real Business Problem (or tackling a live Kaggle competition).
- **Day 02** Data Acquisition: Web Scraping or API Data Extraction.
- **Day 03** Robust Data Pipeline and Feature Store Creation.
- **Day 04** Model Training, Tuning, Experimentation, and SHAP Interpretation.
- **Day 05** Deployment: Creating a Live Dashboard (Streamlit) or API.
- **Day 06** Documentation: Writing a Professional README and Technical Blog Post.
- 🎬 **Day 07** Project Presentation: Recording a 5-Minute Technical Demo.

Week 16: The Job Hunt - Cracking the Interview

- **Day 01** Resume Surgery: Keyword Optimization for ATS.
- **Day 02** LinkedIn Branding: Networking and Cold Outreach Strategies.
- **Day 03** Behavioral Interviews: STAR Method for "Tell me about a time..."
- **Day 04** Technical Case Studies: Translating Business Problems to ML Metrics.
- **Day 05** Mock Technical Interview: ML Theory, Math, and Model Trade-offs.
- **Day 06** Negotiation Skills: Salary, Benefits, and Equity.
- 🏁 **Day 07** Final Review: Revisit Week 01–03 Math (The most common trap).